

Requirements:

1. Create a list or array of historical daily prices that simulates at least 60 days of price movement for a stock.
2. Implement a moving average crossover strategy where:
 - You calculate both a short-term (10-day) and long-term (30-day) moving average
 - A buy signal is generated when the short-term MA crosses above the long-term MA
 - A sell signal is generated when the short-term MA crosses below the long-term MA
3. Use a while loop to simulate moving through the price data day by day, as if you were experiencing it in real-time.
4. Track the following metrics:
 - Entry and exit points (prices and days)
 - Profit/loss for each trade
 - Total number of trades
 - Percentage of winning trades
 - Maximum drawdown (largest drop from a peak)
 - Final portfolio value assuming a starting amount of \$10,000
5. Use conditional statements to handle special cases, such as:
 - Already being in a position when a new buy signal appears
 - Not having a position when a sell signal appears
 - Implementing a stop-loss that exits positions that lose more than 5%